

Oscil - Two-channel "Smart" Oscilloscope and Function Generator

Use-case / Functionality

- Measure voltage
- Measure current
- Output various signals (sine, pulse, sawtooth, etc)
- Save screenshots
 - ↳ locally
 - ↳ cloud
- Download screenshots
 - ↳ through USB
- User account to save screenshots to cloud
- Touchscreen capability (To supplement physical buttons and rotary knobs)
- OTA updates (over Wifi)
- Wifi / Bluetooth

Hardware Requirements

- ADC to convert voltage/amps → ~~Built-in ADC in MCU's ~200 kHz sample rate (Too slow !!!)~~
 - ↳ External ADC, 12-Bit SAR, 10x times sample rate of typical signal
 - (fast and cheap, ~3\$) ✓
- DAC to produce analog signal for function generator → dedicated MCU for DAC → R2R ladder
- 5V Power Supply → needs step down DC to DC converter for 3.3V for ESP32 MCU
 - ↳ also needs boost converter for display
- OTA updates (Wi-Fi) → ESP-32
- MCU choice →
- Touchscreen Big Display → Big Display means → ~~SPI for ST controller for 565 RGB screens (only for small displays)~~
 - ↳ RGB Parallel display → (needs at least 20+ pins
16 Data + 4 sync = 20 pins ") → Dedicated display MCU ✓
 - or → ~~display controller (Expensive ~25-40\$ vs MCU ~5\$)~~

Firmware Requirements

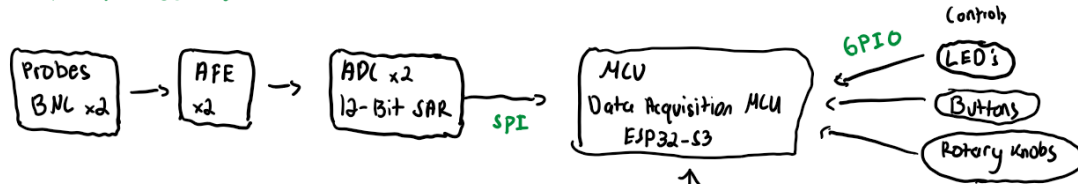
- ESP32 → RTOS (FreeRTOS) → very supported library to reduce error of wasting CPU time with blocking and concurrency
- Data acquisition (from ADC) → SPI
- Display Data → Second dedicated MCU for Display → 20+ parallel pins
- First ESP32 (Data Acquisition) transfer data to Display MCU → Through UART for full duplex communication
- DAC for function generator → Dedicated MCU for generator → I2C ladder → UART communication between Display MCU and Generator MCU
- Wifi for Over-the-air (OTA) firmware updates → flash should be big enough to hold 2 images → At least 8MB or 16MB → ESP32-S3 N16R8 holds 16MB of flash ✓
- Wifi to access cloud database → flash big enough to store wifi libraries
- Low Power → Deep sleep? → ~~will reboot on wake~~ → ~~hinders user experience~~
Design → Light sleep? → doesn't reboot, → still resumes next instruction on wake
Stream still alive → dims backlight and static photo
→ might hinder ADC/DAC operations → feasible, needs further testing
- Continuous Data from ADC → DMA
- Local storage to store user session → NVS so storage persist
- Upload screenshots to USB → ESP32 Native USB Port (Uses UART)
- HTTPS client to access cloud database
- Touch screen interface → coordinates → I2C from screen to MCU Display

Software Requirements

- Database → Supabase
- Security → JSON Web Tokens (JWT)
- Backend Hosting
- User info / User session / Oscilloscope screenshot metadata → Text Data → Supabase Postgres database table
- Oscilloscope screenshots (Jpeg, PNG, wmv, ...etc) → Big Data (BLOB) → Supabase Bucket

Architecture

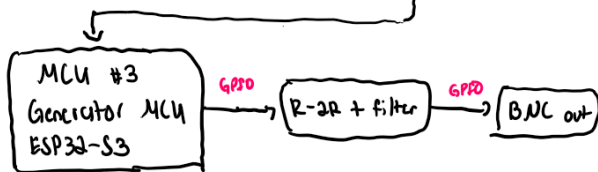
DATA ACQUISITION



DISPLAY



SIGNAL GENERATOR



HTTP

DATABASE / BACKEND SERVER

